

Problem class

(Chapter 1 and Chapter 2)

Chapter 1

1.1 Introduction

1.2 Signal classification and typical signals

1.3 Signal operation

1.4 Singular signals

1.5 System Models and Classification

1.6 Linear Time-Invariant Systems

1.7 System Analysis Methods

Chapter 2

2.1 Introduction

2.2 Establishment of System Mathematical Model (Differential Equation)

2.3 Solving Differential Equations Using the Classical Time-Domain Method

2.4 Jump at the Starting Point

2.5 Zero-Input Response and Zero-State Response

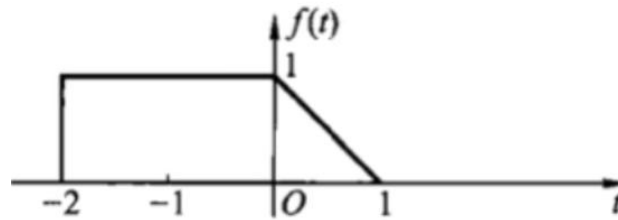
2.6 Impulse Response and Step Response

2.7 Convolution

2.8 Properties of Convolution

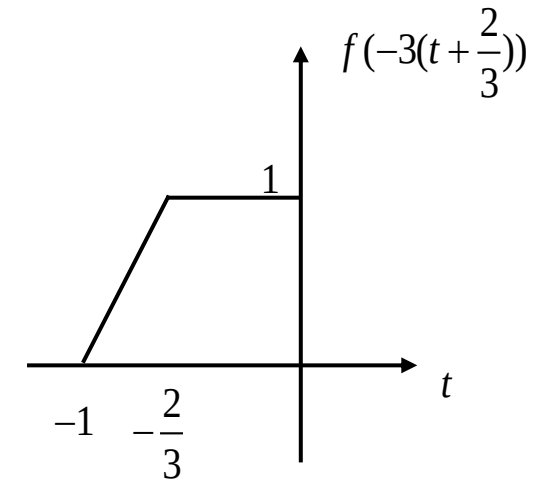
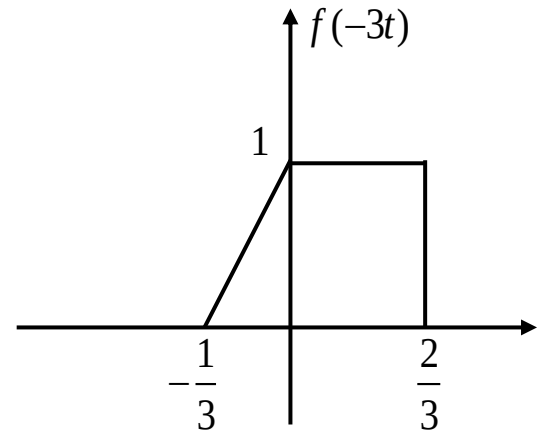
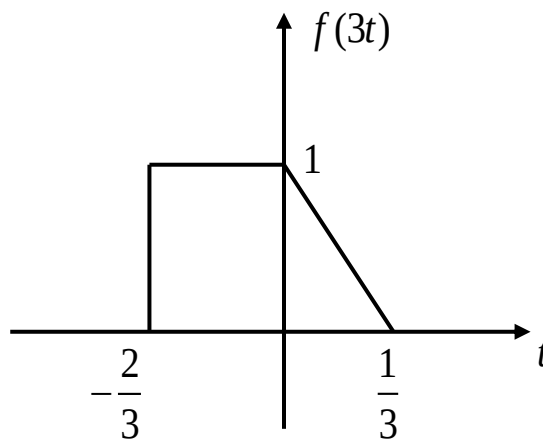
Chapter 1

Problem 1-1: The signal is shown in the figure below, find $f(-3t-2)$, and sketch the waveform.



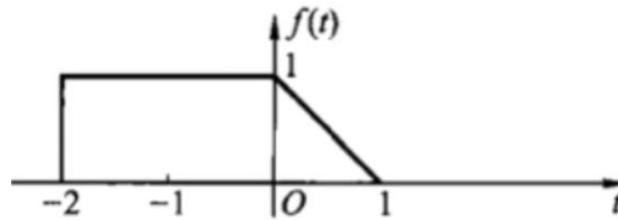
Method 1:

scaling $\rightarrow$ reverse $\rightarrow$ shift



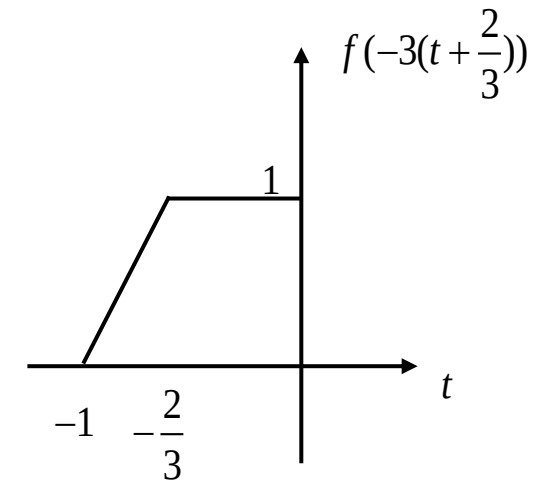
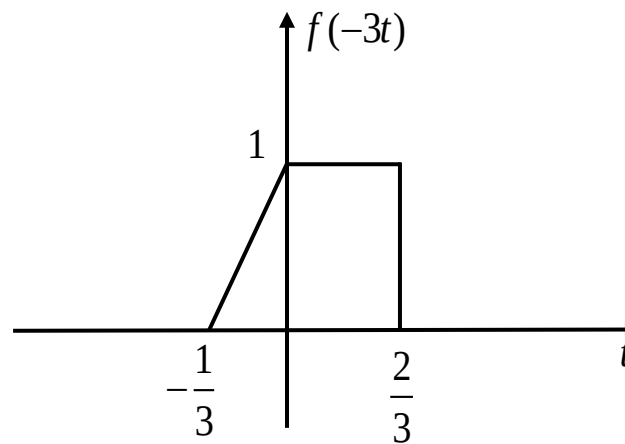
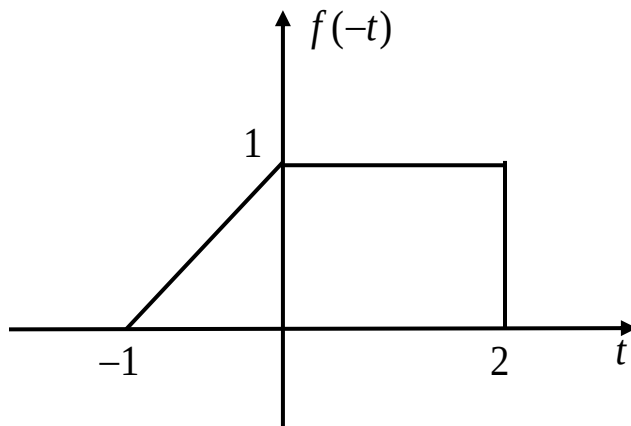
Chapter 1

Problem 1-1: The signal is shown in the figure below, find $f(-3t-2)$, and sketch the waveform.



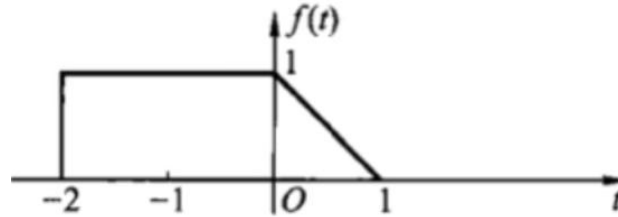
Method 2:

reverse $\rightarrow$ scaling $\rightarrow$ shift



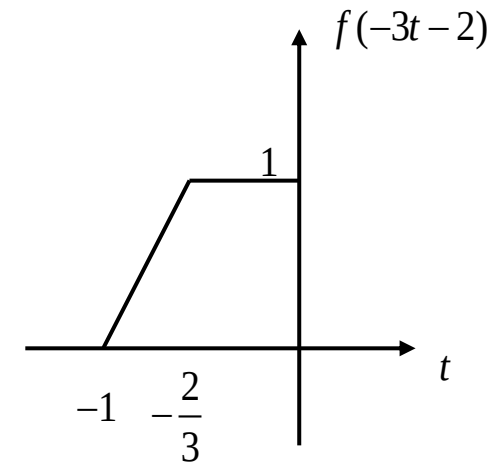
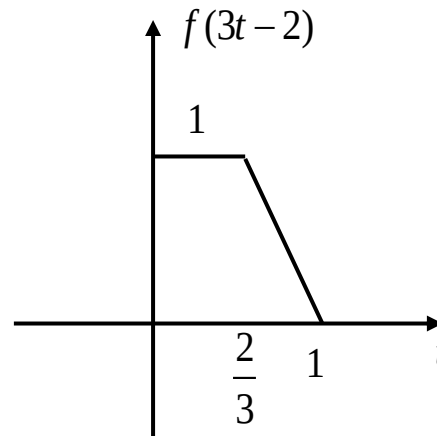
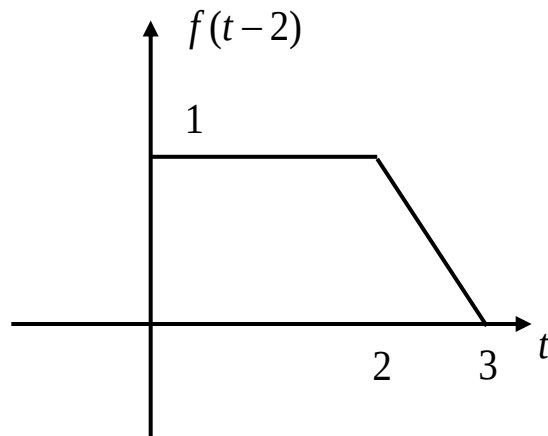
Chapter 1

Problem 1-1: The signal is shown in the figure below, find $f(-3t-2)$, and sketch the waveform.



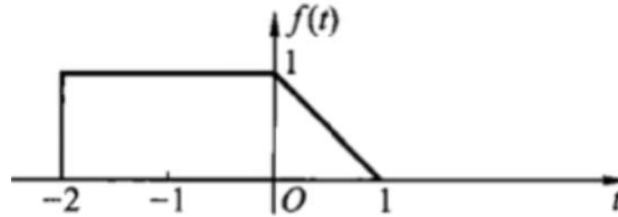
Method 3:

shift $\rightarrow$ scaling $\rightarrow$ reverse



Chapter 1

Problem 1-1: The signal is shown in the figure below, find $f(-3t-2)$, and sketch the waveform.



Solution:

The operation objects of shift, reverse, and scaling operation are **only t** .

Regardless of the operation order, reverse and scaling directly add a negative sign or multiply by the scaling coefficient in front of t , while the shift operation must carry the coefficient in front of t .

Chapter 1

Problem 1-2: Determine the Linearity, Time-Invariance, and Causality of the Following Systems

$$(1) \ r(t) = \sin(e(t))u(t) \quad (2) \ r(t) = e(2t) \quad (3) \ r(t) = \int_{-\infty}^{5t} e(\tau) d\tau$$

Linearity: Check whether the output $r(t)$ for the input $c_1 e_1(t) + c_2 e_2(t)$ is equal to $c_1 r_1(t) + c_2 r_2(t)$

$$(1) \ e_1(t) \rightarrow r_1(t) = \sin(e_1(t))u(t) \ , \ e_2(t) \rightarrow r_2(t) = \sin(e_2(t))u(t)$$

$$c_1 e_1(t) + c_2 e_2(t) \rightarrow r(t) = \sin(c_1 e_1(t) + c_2 e_2(t))u(t) \neq c_1 r_1(t) + c_2 r_2(t)$$

Not linear

$$(2) \ e_1(t) \rightarrow r_1(t) = e_1(2t) \ , \ e_2(t) \rightarrow r_2(t) = e_2(2t)$$

$$c_1 e_1(t) + c_2 e_2(t) \rightarrow r(t) = c_1 e_1(2t) + c_2 e_2(2t) = c_1 r_1(t) + c_2 r_2(t)$$

Linear

$$(3) \ e_1(t) \rightarrow r_1(t) = \int_{-\infty}^{5t} e_1(\tau) d\tau \ , \ e_2(t) \rightarrow r_2(t) = \int_{-\infty}^{5t} e_2(\tau) d\tau$$

$$c_1 e_1(t) + c_2 e_2(t) \rightarrow r(t) = \int_{-\infty}^{5t} c_1 e_1(\tau) + c_2 e_2(\tau) d\tau = c_1 r_1(t) + c_2 r_2(t)$$

Linear

Chapter 1

Problem 1-2: Determine the Linearity, Time-Invariance, and Causality of the Following Systems

$$(1) \ r(t) = \sin(e(t))u(t) \quad (2) \ r(t) = e(2t) \quad (3) \ r(t) = \int_{-\infty}^{5t} e(\tau) d\tau$$

Time-Invariance: Check whether the output $r(t)$ for the input $e(t-t_0)$ is equal to $r(t-t_0)$

$$(1) \ e(t-t_0) \rightarrow r(t) = \sin(e(t-t_0))u(t) \neq \sin(e(t-t_0))u(t-t_0) = r(t-t_0) \quad \text{Time varying}$$

$$(2) \ e(t-t_0) \rightarrow r(t) = e(2t-t_0) \neq e(2t-2t_0) = r(t-t_0) \quad \text{Time varying}$$

$$(3) \ e(t-t_0) \rightarrow r(t) = \int_{-\infty}^{5t} e(\tau-t_0) d\tau = \int_{-\infty}^{5t-t_0} e(\tau) d\tau \neq \int_{-\infty}^{5t-5t_0} e(\tau) d\tau = r(t-t_0) \quad \text{Time varying}$$

Summary: If there is a time-varying coefficient before $e(\cdot)$, such as $r(t) = te(t)$; or if there are scaling transformations of $e(t)$, such as $r(t) = e(2t)$, $r(t) = e(-t)$, $r(t) = e(1-t)$; or if $r(t)$ contains $u(t)$, such as $r(t) = e(\cdot)u(t)$, the system is a time-varying system.

Chapter 1

Problem 1-2: Determine the Linearity, Time-Invariance, and Causality of the Following Systems

$$(1) \ r(t) = \sin(e(t))u(t) \quad (2) \ r(t) = e(2t) \quad (3) \ r(t) = \int_{-\infty}^{5t} e(\tau) d\tau$$

Causality : The output at the current moment **depends only on the input at the current moment or previous moments.**

(1) Causal

(2) Non Causal

(3) Non Causal

Tips: If the function inside $e(\cdot)$ is greater than t in $r(t)$ (which can be judged by drawing graphs), such as $r(t) = e(2t)$, $r(t) = e(1 - t)$, then the system is a non-causal system; otherwise, it is a causal system.

Chapter 2

Review:

- The complete response of a system solved by the classical method can be divided into:

Complete Response = Free Response + Forced Response

$$r(t) = r_h(t) + r_p(t)$$

- The complete response of the system can also be divided into:

Complete Response = Zero-Input Response + Zero-State Response

$$r(t) = r_{zi}(t) + r_{zs}(t)$$

Suppose the differential equation of an LTI system is $r'(t) + r(t) = e(t)$, and the complete response is $r(t) = \left(1 + \frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t}\right)u(t)$. Then the forced response is ()

- ☐ A $e^{-t}u(t)$
- ☐ B $\frac{1}{2}e^{-t}u(t)$
- ☒ C $(1 - \frac{1}{2}e^{-3t})u(t)$
- ☐ D $(1 + \frac{1}{2}e^{-t})u(t)$

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Chapter 2

Problem 2-1:

Suppose the differential equation of an LTI system is $r'(t) + r(t) = e(t)$, and the complete response is $r(t) = \left(1 + \frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t}\right)u(t)$. Find the forced response.

Solution:

The free response is only related to the homogeneous solution.

The characteristic equation of the system is $\alpha + 1 = 0$, and the characteristic root is $\alpha = -1$. Therefore, the free response of the system is $\frac{1}{2}e^{-t}u(t)$.

The forced response is obtained by subtracting the free response from the complete response : $\left(1 - \frac{1}{2}e^{-3t}\right)u(t)$.

Problem 2-2: Suppose the differential equation of a linear time-invariant system is

$$\frac{d^2 r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = \frac{de(t)}{dt} + 3e(t)$$

where $e(t)$ is the excitation and $r(t)$ is the response. Given $e(t) = e^{-t}u(t)$, the complete response of the system is $r(t) = [(2t + 3)e^{-t} - 2e^{-2t}]u(t)$.

- (1) Directly distinguish the forced response and the free response of the system, and give the explanation.
- (2) Find the initial states $r(0_-)$ and $r'(0_-)$ of the system using the impulse function matching method.
- (3) Find the zero-input response and zero-state response of the system.

Chapter 2

Solution:

(1) The free response is only related to the homogeneous solution.

The characteristic equation of the system is $\alpha^2 + 3\alpha + 2 = 0$

The characteristic roots are: $\alpha_1 = -1, \alpha_2 = -2$

Therefore, the free response (homogeneous solution) of the system is

$$r_h = (3e^{-t} - 2e^{-2t})u(t)$$

Subtracting the free response from the complete response gives the forced response as

$$\begin{aligned} r_p(t) &= r(t) - r_h(t) \\ &= [(2t + 3)e^{-t} - 2e^{-2t}]u(t) - (3e^{-t} - 2e^{-2t})u(t) \\ &= 2te^{-t}u(t) \end{aligned}$$

(The excitation $e(t) = e^{-t}u(t)$ contains a characteristic root -1.)

Chapter 2

(2) When $e(t) = e^{-t}u(t)$, $\frac{d^2r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = e^{-t}\delta(t) - e^{-t}u(t) + 3e^{-t}u(t)$

$$\frac{d^2r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = \delta(t) + 2e^{-t}u(t)$$

When $t=0$, $\frac{d^2r(t)}{dt^2} + 3\frac{dr(t)}{dt} + 2r(t) = \delta(t) + 2\Delta u(t)$

By the impulse function matching method, it is known that **$r'(t)$ has a jump of +1 from 0_- to 0_+ , while $r(t)$ does not change from 0_- to 0_+ .**

$$\frac{d^2r(t)}{dt^2} = \delta(t), \quad \frac{dr(t)}{dt} = 1 \cdot \Delta u(t), \quad r(t) = 0$$

$\therefore r(0_+) = r(0_-), \quad r'(0_+) - r'(0_-) = 1$ **State jump amount from 0_- to 0_+**

$$r(t) = [(2t + 3)e^{-t} - 2e^{-2t}]u(t), \quad r(0_+) = 1$$

$$r'(t) = [2e^{-t} - (2t + 3)e^{-t} + 4e^{-2t}]u(t) = [-(2t + 1)e^{-t} + 4e^{-2t}]u(t), \quad r'(0_+) = 3$$

$$r(0_-) = r(0_+) = 1, \quad r'(0_-) = r'(0_+) - 1 = 3 - 1 = 2$$

Chapter 2

(3) Finding the zero-input and zero-state responses.

Method 1: Find the zero-input response first.

Characteristic roots of the system: $\alpha_1 = -1, \alpha_2 = -2$

Zero-input response: $r_{zi}(t) = A_1 e^{-t} + A_2 e^{-2t} \quad (t > 0)$

$$r'_{zi}(t) = -A_1 e^{-t} - 2A_2 e^{-2t} \quad (t > 0)$$

Substitute the initial states, $r_{zi}(0_+) = r_{zi}(0_-) = r(0_-) = 1,$

$$r'_{zi}(0_+) = r'_{zi}(0_-) = r'(0_-) = 2$$

$$\begin{cases} A_1 + A_2 = 1 \\ -A_1 - 2A_2 = 2 \end{cases} \quad \longrightarrow \quad A_1 = 4, A_2 = -3$$

So the zero-input response is $r_{zi}(t) = (4e^{-t} - 3e^{-2t})u(t)$

The zero-state response is the complete response minus the zero-input response:

$$r_{zs}(t) = r(t) - r_{zi}(t) = [(2t - 1)e^{-t} + e^{-2t}]u(t)$$

Chapter 2

Method 2: Find the zero-state response first.

$$r_{zs}(t) = \underbrace{(A_{zs1}e^{-t} + A_{zs2}e^{-2t})u(t)}_{\text{homogeneous solution}} + \underbrace{Bte^{-t}u(t)}_{\text{particular solution}}$$

The particular solution is the forced response. It was solved in (1) as $B=2$.

Alternatively, the particular solution and the excitation can be substituted into the left and right sides of the differential equation respectively to find $B = 2$.

It is known from (2) that that $r(t)$ does not change from 0_- to 0_+ , while $r'(t)$ changes with an increment of 1.

$$\therefore r_{zs}(0_+) = r(0_+) - r(0_-) = 0, \quad r'_{zs}(0_+) = r'(0_+) - r'(0_-) = 1$$

$$r_{zs}(0_+) = A_{zs1} + A_{zs2} = 0 \quad \rightarrow A_{zs1} = -1, \quad A_{zs2} = 1.$$

$$r'_{zs}(0_+) = -A_{zs1} - 2A_{zs2} + 2 = 1$$

$$r_{zs}(t) = (2te^{-t} - e^{-t} + e^{-2t})u(t)$$

The zero-input response is the complete response minus the zero-state response:

$$r_{zi}(t) = (4e^{-t} - 3e^{-2t})u(t)$$

Given the following differential equation of an LTI system, the impulse response $h(t)$ is () 。

$$\frac{dr(t)}{dt} + 2r(t) = 2e(t)$$

A $h(t) = -2e^{-2t}u(t)$

B $h(t) = -e^{-2t}u(t)$

C $h(t) = 2e^{-2t}u(t)$

D $h(t) = e^{-2t}u(t),$

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Chapter 2

Problem 2-3: Given the following differential equation of an LTI system, Find impulse response $h(t)$ and step response $g(t)$.

$$\frac{dr(t)}{dt} + 2r(t) = 2e(t)$$

Solution:

1. Find Impulse Response $h(t)$

Step 1: Set excitation $e(t) = \delta(t)$, system equation becomes $\frac{dh(t)}{dt} + 2h(t) = 2\delta(t)$

Step 2: Analyze jumps: $\frac{dh(t)}{dt} = 2\delta(t)$, so $h(t) = 2\Delta u(t)$ Jump value of $h(t)$ from 0_- to 0_+ is 2, $h(0_+) = 2$.

Step 3: Solve homogeneous equation: obtain characteristic root $\alpha_1 = -2$

Impulse Response : $h(t) = 2e^{\alpha_1 t} u(t) = 2e^{-2t} u(t)$

$h(0_+) = 2$ is the initial value, and the characteristic root determines the speed of signal attenuation. When there is only one characteristic root, it can be directly written in this form.

Given waveforms of $f_1(t)$ and $f_2(t)$.

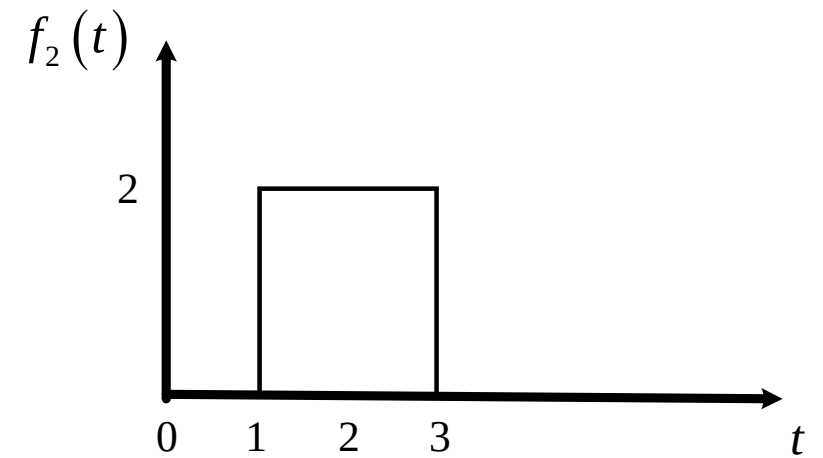
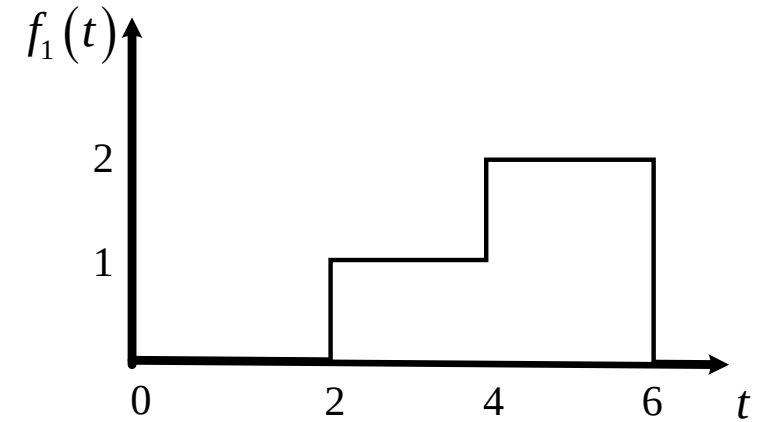
If $s(t) = f_1(t) * f_2(t)$, find the value $s(6)$.

A 2

B 3

C 4

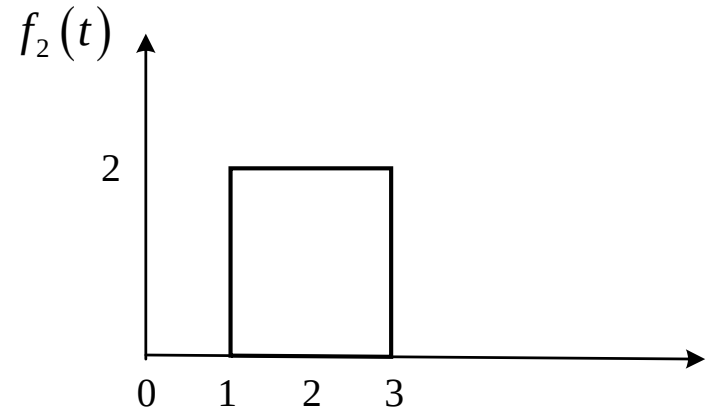
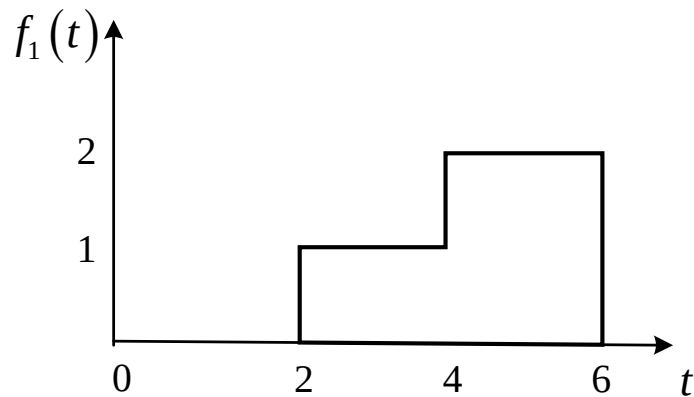
D 6



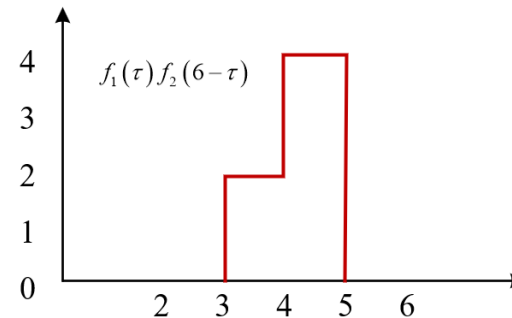
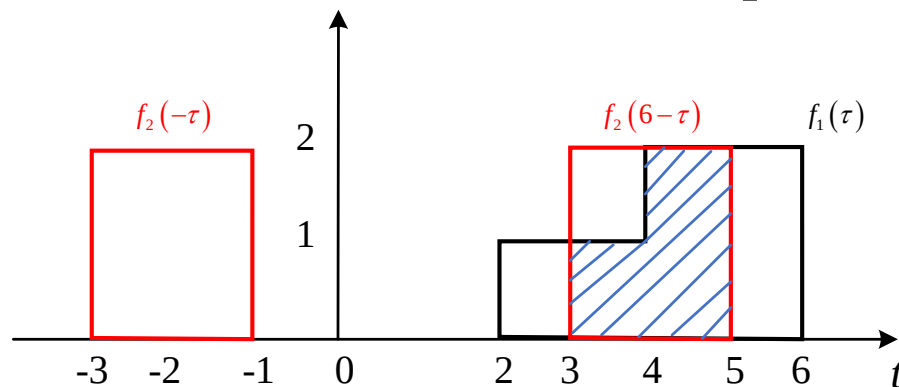
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Chapter 2

Problem 2-4: Given waveforms of $f_1(t)$ and $f_2(t)$. If $s(t) = f_1(t) * f_2(t)$, find $s(6)$.



- Graphical Method:**
1. Flip: Flip $f_2(\tau)$ along the τ axis to get $f_2(-\tau)$.
 2. Shift: Shift $f_2(-\tau)$ to the right by $t=6$ to get $f_2(6-\tau)$.
 3. Overlap Analysis: Find the overlapping interval of $f_1(\tau)$ and $f_2(6-\tau)$.



$$s(6) = \int_{-\infty}^{\infty} f_1(\tau) f_2(6-\tau) d\tau$$

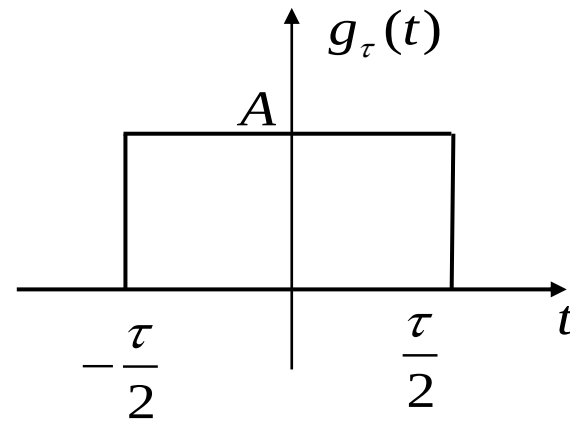
$$s(6) = 2 \times 1 \times 1 + 2 \times 2 \times 1 = 6$$

The graphical method is suitable for finite waveforms and is more convenient for calculating the convolution value at a specific moment.

Chapter 2

Problem 2-5:

For two rectangular pulses $g_\tau(t)$, with pulse widths τ , and amplitudes A , find the convolution $g_\tau(t)*g_\tau(t)$.



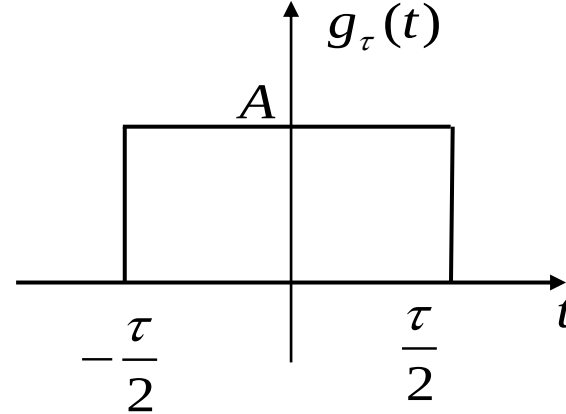
Solution:

For rectangular pulses $g_\tau(t)$:

$g_\tau(t)=A$ when $-\frac{\tau}{2} \leq t \leq \frac{\tau}{2}$, and 0 otherwise.

$g_\tau(t-\tau)=A$ when $t-\frac{\tau}{2} \leq \tau \leq t+\frac{\tau}{2}$, and 0 otherwise.

Chapter 2

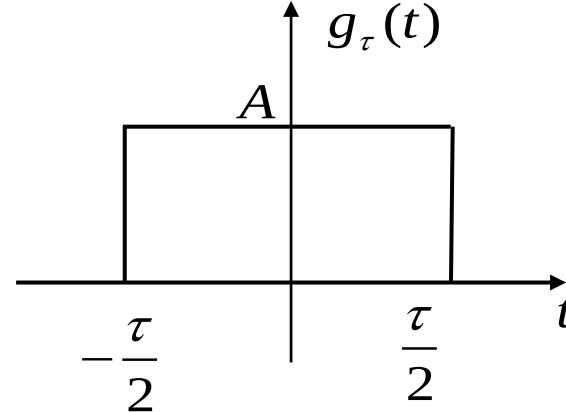


Case 1: $t \leq -\frac{\tau + \tau}{2}$ No overlap: $g_\tau(t) g_\tau(t - \tau) = 0$. $g_\tau(t) * g_\tau(t) = 0$

Case 2: $-\frac{\tau + \tau}{2} < t \leq 0$ Overlap interval: $-\frac{\tau}{2} \leq \tau \leq t + \frac{\tau}{2}$

$$g_\tau(t) * g_\tau(t) = \int_{-\frac{\tau}{2}}^{t + \frac{\tau}{2}} A \cdot A d\tau = A^2(t + \tau)$$

Case 3: $t = 0$ Maximum overlap region $g_\tau(t) * g_\tau(t) = A^2\tau$



Case 4: $0 < t \leq \frac{\tau + \tau}{2}$ Overlap interval: $t - \frac{\tau}{2} \leq \tau \leq \frac{\tau}{2}$

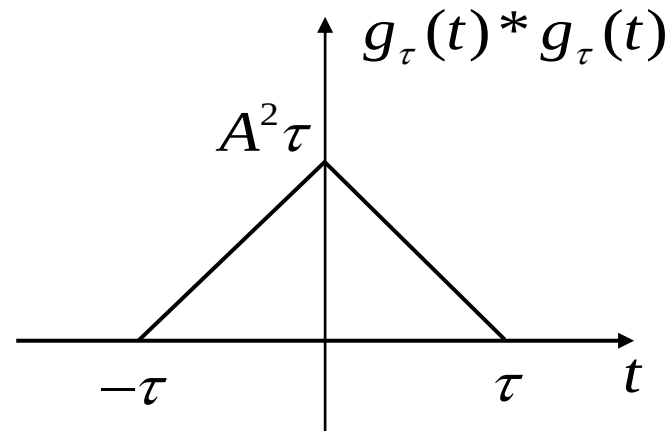
$$g_\tau(t) * g_\tau(t) = \int_{t-\frac{\tau}{2}}^{\frac{\tau}{2}} A \cdot A d\tau = A^2(\tau - t)$$

Case 5: $\frac{\tau + \tau}{2} < t$ No overlap: $g_\tau(t) g_\tau(t - \tau) = 0$. $g_\tau(t) * g_\tau(t) = 0$

Chapter 2

Rectangular pulses convolve to produce triangular pulses.

For two rectangular pulses $g_\tau(t)$, with pulse widths τ , and amplitudes A , find the convolution $g_\tau(t) * g_\tau(t)$.



- The convolution result of **two rectangular pulses with equal widths** is a **triangular function**. The **height is the product of the two rectangular heights and the width of one rectangle**, and the **lower base is the sum of the two rectangular widths** (twice the width of one rectangle).

For a certain LTI system, when the excitation is $f(t)$, the impulse response of the system is $h(t)$, the zero-state response is $y_{zs}(t)$, and the zero-input response is $y_{zi}(t)$. If the initial state remains unchanged and the input excitation changes to $2f(t)$, the total response of the system is:

- ☒ A $y_{zi}(t) + 2y_{zs}(t)$
- ☒ B $y_{zi}(t) + 2f(t) * h(t)$
- ☐ C $y_{zs}(t) + 2y_{zi}(t)$
- ☐ D $y_{zs}(t) + 2f(t) * h(t)$

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Chapter 2

Problem 2-6: For an LTI system, when the excitation is $f(t)$, the impulse response of the system is $h(t)$, the zero state response is $y_{zs}(t)$, and the zero input response is $y_{zi}(t)$. If the initial state remains unchanged and the input excitation changes to $2f(t)$, find the total response of the system.

Solution:

The zero input response is only related to the initial state of the system. Since the initial state of the system remains unchanged, the zero - input response for the second time is still $y_{zi}(t)$.

The zero state response of the system can be obtained by convolving the input signal with the impulse response: $2f(t) * h(t)$.

According to the properties of linear systems, it can also be expressed as $2y_{zs}(t)$.

Therefore, the total response corresponding to the second input can be expressed as: $y_{zi}(t) + 2y_{zs}(t)$ or $y_{zi}(t) + 2f(t) * h(t)$.

Given $f(t) = u(t - 1) + u(t - 3)$, $x(t) = \delta(t - 3)$, find $f(t) * x(t) = ?$

- ☐ A $\delta(t - 4) + \delta(t - 6)$
- ☐ B $u(t - 1) + u(t - 3)$
- ☒ C $u(t - 4) + u(t - 6)$
- ☐ D $\delta(t - 1) + \delta(t - 3)$

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Solving Each Response Component in Time Domain

- Establishment of system differential equation:

Relevant knowledge of circuits:

(1) KCL and KVL equations;

(2) The voltage across the capacitor and the current across the inductor **cannot change suddenly** (commonly used for calculating boundary conditions $r(0_-)$ and $r(0_+)$);

Voltage - current relationships of capacitors and inductors:

$$i_C = C \frac{du_C}{dt} \qquad u_L = L \frac{di_L}{dt}$$

- Complete response = homogeneous solution (free response) + particular solution (forced response) = zero - input response + zero - state response

Summary

Solving Each Response Component by Time Domain Method

Step 1: Write the differential equation describing the system.

Step 2: Solve the response:

① The form of the **homogeneous solution (free response)** $r_h(t)$ is determined by the **characteristic root**, and the form of the **particular solution (forced response)** $r_p(t)$ is determined by the **free term**;

Method for determining coefficients in the particular solution: Substitute the particular solution with coefficients into the differential equation to find the coefficients;

Method for determining coefficients in the homogeneous solution:

Complete response = homogeneous solution with coefficients + obtained particular solution

Substituting the **boundary conditions at time 0_+** , find the coefficients of the homogeneous solution, and obtain the complete response.

Solving Each Response Component by Time Domain Method

Step 2: Solve the response:

② The form of the **zero - input response** $r_{zi}(t)$ (same as the form of the homogeneous solution) is determined by the **characteristic root**. Directly substitute the **boundary conditions at time 0_-** to find the zero - input response.

The **zero - state response** can be obtained by two method:

Method 1: Convolution

Calculate the impulse response $h(t)$ and then find the zero - state response through convolution:

$$r_{zs}(t) = h(t) * e(t)$$

Method 2: homogeneous solution + particular solution $r_{zs}(t) = r_{zsh}(t) + r_p(t)$

Coefficients in the homogeneous solution of zero-state response can be obtained by **jump value from 0_- to 0_+** .

Calculation of Convolution

- Formula method: $f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$
- Graphical method: flip, shift, multiply, and sum.
- Application of convolution properties:

Differentiation and integration of convolution:

$$\frac{d}{dt} [f_1(t) * f_2(t)] = f_1(t) * \frac{df_2(t)}{dt} = \frac{df_1(t)}{dt} * f_2(t)$$

$$\int_{-\infty}^t f_1(\lambda) f_2(\lambda) d\lambda = f_1(\lambda) * \int_{-\infty}^t f_2(\lambda) d\lambda = f_2(\lambda) * \int_{-\infty}^t f_1(\lambda) d\lambda$$

Convolution with impulse function or step function:

$$f(t) * \delta(t) = f(t) \qquad f(t) * \delta(t - t_0) = f(t - t_0)$$

$$f(t) * \delta'(t) = f'(t) \qquad f(t) * u(t) = \int_{-\infty}^t f(\tau) d\tau$$